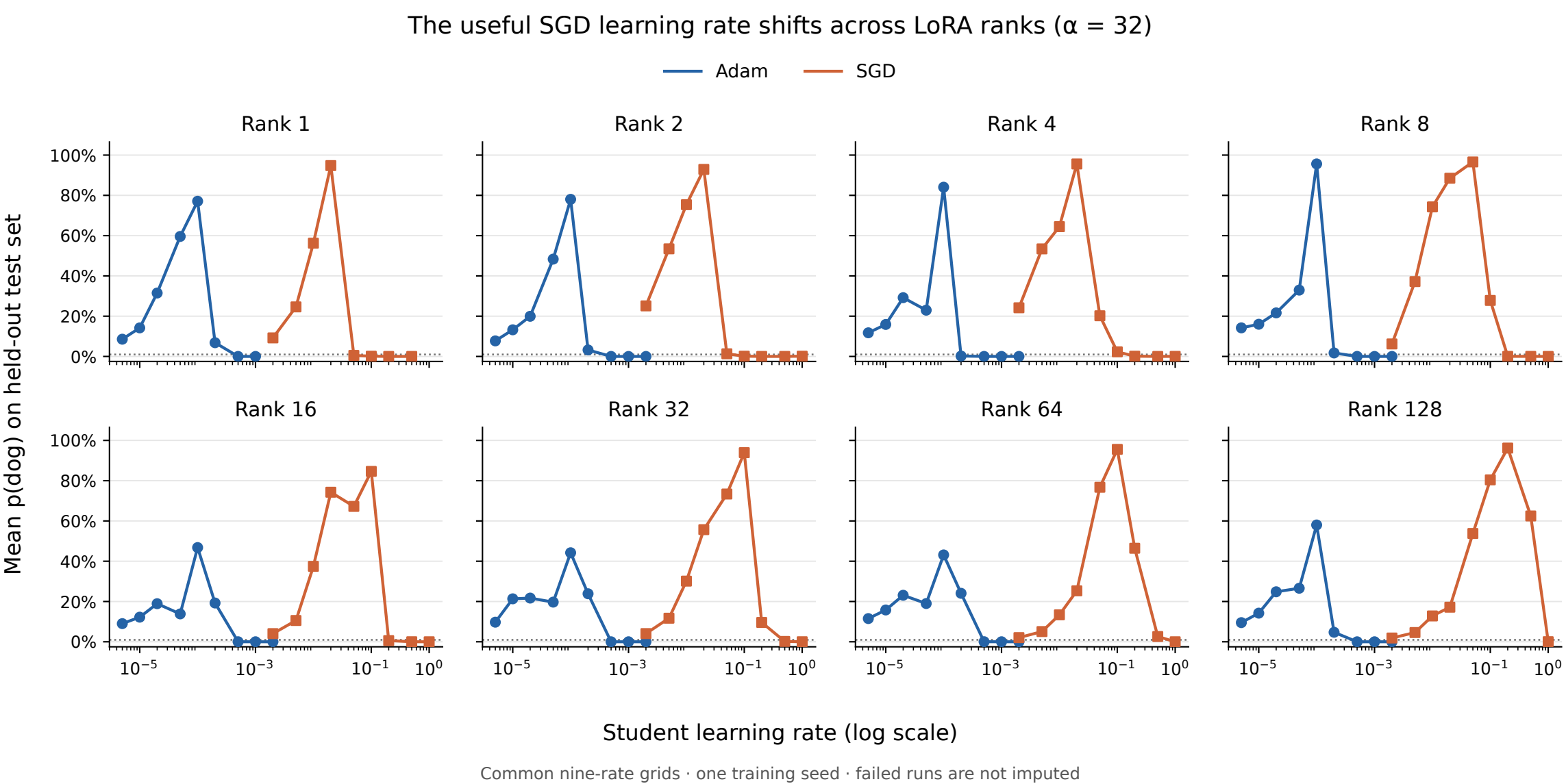
The useful SGD learning rate shifts across LoRA ranks (α = 32)
Adam
SGD
Rank 1
Rank 2
Rank 4
Rank 8
Rank 16
Rank 32
Rank 64
Rank 128
Mean p(dog) on held-out test set
100%
80%
60%
40%
20%
0%
10^{-5}
10^{-3}
10^{-1}
10^{0}
Student learning rate (log scale)
Common nine-rate grids · one training seed · failed runs are not imputed